

Introduction to Social Data Analytics: Week 1 Part 2

Overview of Lecture

- ▶ Using Statistical functions
- ▶ The impact of providing financial incentives to instructors
- ▶ Pivot tables

Illustrative Example

- ▶ Let's consider the following dataset that has information on Gross Domestic Product (GDP) for ten countries

	A	B	C	D	E	F	G	H	I
1	Country	Year	GDP (trillions)						
2	United States	2020	20.94						
3	China	2020	14.72						
4	Japan	2020	5.06						
5	Germany	2020	3.81						
6	United Kingdom	2020	2.71						
7	India	2020	2.62						
8	France	2020	2.6						
9	Italy	2020	1.89						
10	Canada	2020	1.64						
11	South Korea	2020	1.63						
12									
13									
14									
15									

Summary Statistics

- ▶ Often we want to retrieve simple summary statistics of our dataset
 - ▶ What is the number of observations, the average value of a variable, etc.
- ▶ To retrieve these statistics in Excel we use **statistical functions**

Functions

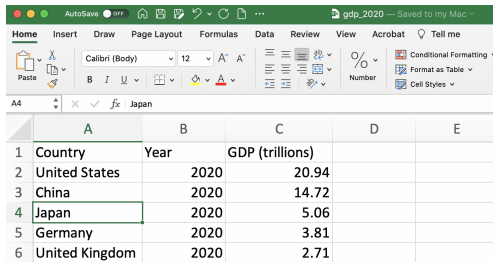
- ▶ In general, a **function** takes in an input and produces some output
- ▶ In Excel, **statistical functions** take in a number or set of numbers, and outputs some function of these numbers
- ▶ For example, the AVERAGE function will take in a list of numbers and output the average of all those numbers

Examples of Statistical Functions

- ▶ Examples of Statistical functions
 - ▶ COUNT – counts the number of numbers
 - ▶ SUM – adds up the numbers
 - ▶ AVERAGE – averages the numbers
 - ▶ MEDIAN – retrieves the median of all numbers
 - ▶ MAX, MIN – retrieves the maximum and minimum of all numbers respectively
 - ▶ MODE – retrieves the mode of all numbers

Referencing Cells

- ▶ In order to make use of statistical functions, we need to learn how to reference cells
- ▶ Every cell in Excel is identifiable by the column letter and row number
- ▶ For example, in the dataset below, the entry of A4 (column A row 4) is Japan

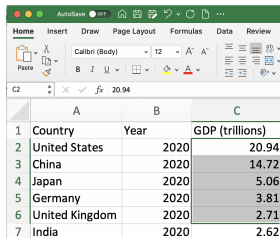


The screenshot shows the Microsoft Excel interface with the 'gdp_2020' file open. The 'Home' tab is selected, and the ribbon shows various formatting options. The active cell is A4, which contains the text 'Japan'. The formula bar shows 'Japan'. The worksheet contains a table with the following data:

	A	B	C	D	E
1	Country	Year	GDP (trillions)		
2	United States	2020	20.94		
3	China	2020	14.72		
4	Japan	2020	5.06		
5	Germany	2020	3.81		
6	United Kingdom	2020	2.71		

Referencing a range of cells

- ▶ You can also reference a range of cells (this will be useful when taking averages)
- ▶ For example, to reference rows 2-6 of column C we could type C2:C6
 - ▶ We read the colon here as “through”
 - ▶ So this reads “cells C2 through C6”



	A	B	C
1	Country	Year	GDP (trillions)
2	United States	2020	20.94
3	China	2020	14.72
4	Japan	2020	5.06
5	Germany	2020	3.81
6	United Kingdom	2020	2.71
7	India	2020	2.62

Referencing an entire column

- ▶ To reference an entire column (for example column C) we can type C:C
 - ▶ This helps if your excel spreadsheet is very long and you don't know how many observations there are

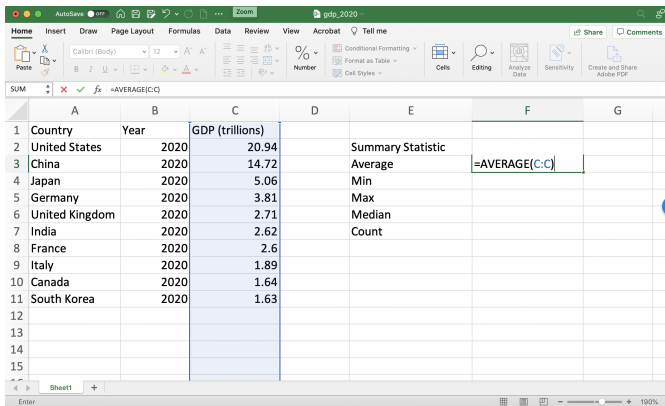
Statistical Functions

- ▶ In order to use a statistical function you need to specify (1) the function you are using and (2) the range of cells to apply it to
- ▶ In our example, if we want to populate a blank cell with the average GDP across all countries, we can type

```
=AVERAGE(C:C)
```

- ▶ Note the = sign. This tells excel we want the value of the cell to be equal to the average of column C

Average GDP



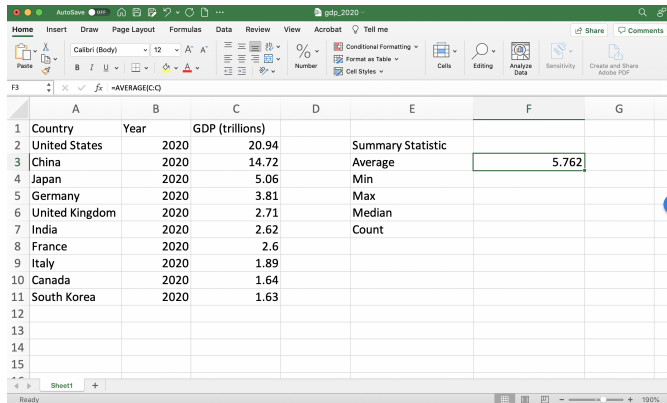
The screenshot shows a Microsoft Excel spreadsheet titled 'gdp_2020'. The spreadsheet has columns A through G and rows 1 through 15. The data is organized as follows:

	A	B	C	D	E	F	G
1	Country	Year	GDP (trillions)				
2	United States	2020	20.94		Summary Statistic		
3	China	2020	14.72		Average	=AVERAGE(C:C)	
4	Japan	2020	5.06		Min		
5	Germany	2020	3.81		Max		
6	United Kingdom	2020	2.71		Median		
7	India	2020	2.62		Count		
8	France	2020	2.6				
9	Italy	2020	1.89				
10	Canada	2020	1.64				
11	South Korea	2020	1.63				
12							
13							
14							
15							

The formula bar at the top shows the formula `=AVERAGE(C:C)` being entered into cell F3. The status bar at the bottom indicates the current cell is F3 and the formula is being entered.

- ▶ After typing in the function, press enter to have the cell fill with the correct value

Average GDP



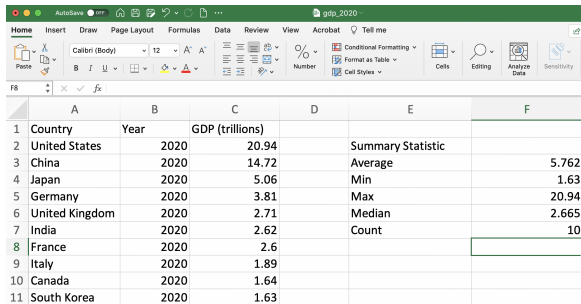
The screenshot shows an Excel spreadsheet titled 'gdp_2020'. The data is organized as follows:

	A	B	C	D	E	F	G
1	Country	Year	GDP (trillions)				
2	United States	2020	20.94		Summary Statistic		
3	China	2020	14.72		Average	5.762	
4	Japan	2020	5.06		Min		
5	Germany	2020	3.81		Max		
6	United Kingdom	2020	2.71		Median		
7	India	2020	2.62		Count		
8	France	2020	2.6				
9	Italy	2020	1.89				
10	Canada	2020	1.64				
11	South Korea	2020	1.63				
12							
13							
14							
15							

The formula bar at the top shows the formula `=AVERAGE(C:C)` and the status bar at the bottom indicates 'Ready'.

- Note: if you forget the equals sign the cell will just read “AVERAGE(C:C)” but not be replaced by the actual value. The equals sign can be easy to forget!

Summary Statistics



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F
1	Country	Year	GDP (trillions)			
2	United States	2020	20.94		Summary Statistic	
3	China	2020	14.72		Average	5.762
4	Japan	2020	5.06		Min	1.63
5	Germany	2020	3.81		Max	20.94
6	United Kingdom	2020	2.71		Median	2.665
7	India	2020	2.62		Count	10
8	France	2020	2.6			
9	Italy	2020	1.89			
10	Canada	2020	1.64			
11	South Korea	2020	1.63			

- The rest of the summary statistics are completely analogous, replacing AVERAGE with the relevant function

The IF function

- ▶ The IF function is useful for creating new variables in Excel, and is an example of a logical function
- ▶ The way it works is that if a statement is true, the function returns a certain value, if it is false, it returns a different value

```
=IF(logical test,value if TRUE, value if FALSE)
```

Logical Statements

- ▶ A logical statement is a statement that is either true or false
 - ▶ We will discuss this in more depth when we get to R or Stata
- ▶ This is sometimes referred to as Boolean logic
 - ▶ Boolean algebra is the branch of mathematics that deals with variables in which the values are either true or false. It is an essential branch of math for the development of computers

Creating Binary Variables

- ▶ In our Brownback and Sadoff (2020) data, we can use the IF to create a new variable that is equal to 1 if the student passes the test (score above 70) and zero otherwise
- ▶ **First Step:** Title the column with the variable name “Passed”
- ▶ **Second Step:** Use the IF function to fill in this variable for the first observation
- ▶ **Third Step:** Double click bottom right of cell in step 2 to fill in the variable for all observations

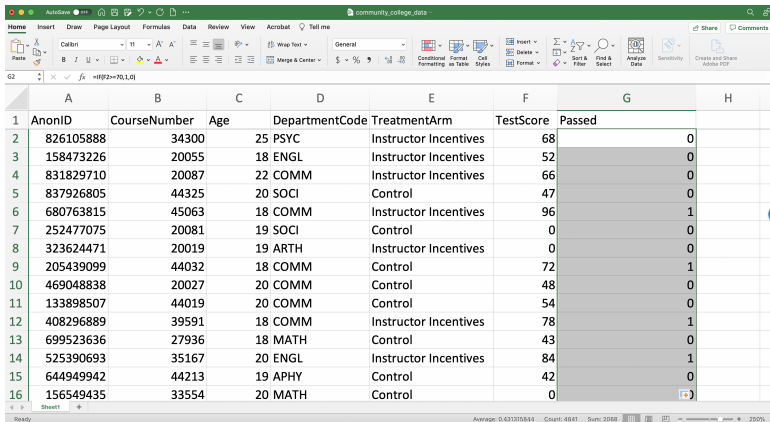
First and Second Step

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H
1	AnonID	CourseNumber	Age	DepartmentCode	TreatmentArm	TestScore	Passed	
2	826105888	34300	25	PSYC	Instructor Incentives	68	=IF(F2>=70,1,0)	
3	158473226	20055	18	ENGL	Instructor Incentives	52		
4	831829710	20087	22	COMM	Instructor Incentives	66		
5	837926805	44325	20	SOCI	Control	47		
6	680763815	45063	18	COMM	Instructor Incentives	96		
7	252477075	20081	19	SOCI	Control	0		
8	323624471	20019	19	ARTH	Instructor Incentives	0		
9	205439099	44032	18	COMM	Control	72		
10	469048838	20027	20	COMM	Control	48		
11	133898507	44019	20	COMM	Control	54		
12	408296889	39591	18	COMM	Instructor Incentives	78		
13	699523636	27936	18	MATH	Control	43		
14	525390693	35167	20	ENGL	Instructor Incentives	84		
15	644949942	44213	19	APHY	Control	42		
16	156549435	33554	20	MATH	Control	0		

Third Step

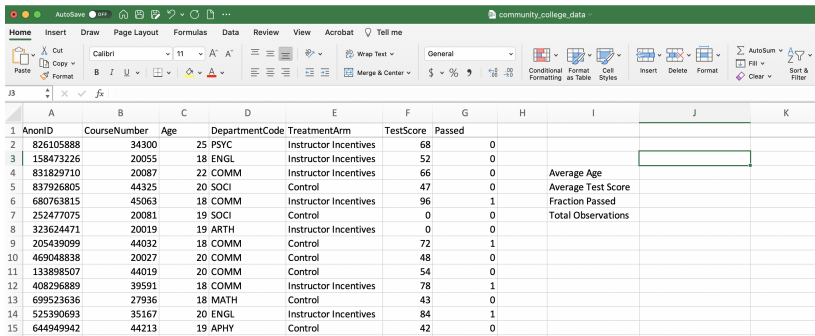
- Double click bottom right of G2 to fill in the rest of the observations



	A	B	C	D	E	F	G	H
1	AnonID	CourseNumber	Age	DepartmentCode	TreatmentArm	TestScore	Passed	
2	826105888	34300	25	PSYC	Instructor Incentives	68	0	
3	158473226	20055	18	ENGL	Instructor Incentives	52	0	
4	831829710	20087	22	COMM	Instructor Incentives	66	0	
5	837926805	44325	20	SOCI	Control	47	0	
6	680763815	45063	18	COMM	Instructor Incentives	96	1	
7	252477075	20081	19	SOCI	Control	0	0	
8	323624471	20019	19	ARTH	Instructor Incentives	0	0	
9	205439099	44032	18	COMM	Control	72	1	
10	469048838	20027	20	COMM	Control	48	0	
11	133898507	44019	20	COMM	Control	54	0	
12	408296889	39591	18	COMM	Instructor Incentives	78	1	
13	699523636	27936	18	MATH	Control	43	0	
14	525390693	35167	20	ENGL	Instructor Incentives	84	1	
15	644949942	44213	19	APHY	Control	42	0	
16	156549435	33554	20	MATH	Control	0		

Summary Statistics

- Now that we've generated the variables we need let's fill in some summary statistics displayed on the right side of the datasheet

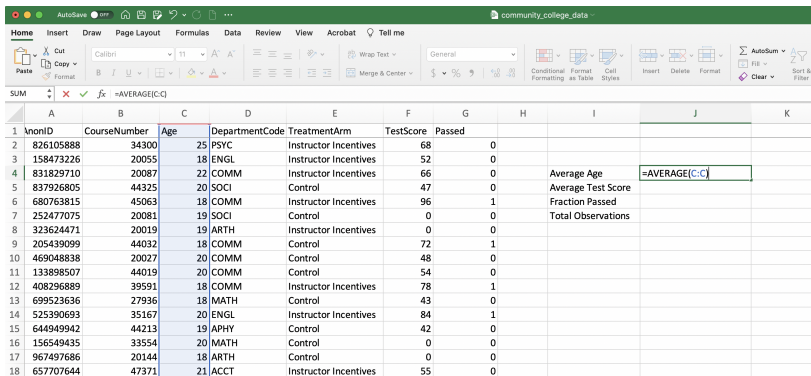


The screenshot shows a Microsoft Excel spreadsheet titled "community_college_data". The ribbon is set to "Home". The spreadsheet contains a table with 15 rows of data. The columns are labeled A through K. The data is as follows:

	A	B	C	D	E	F	G	H	I	J	K
1	AnonID	CourseNumber	Age	DepartmentCode	TreatmentArm	TestScore	Passed				
2	826105888	34300	25	PSYC	Instructor Incentives	68	0				
3	158473226	20055	18	ENGL	Instructor Incentives	52	0				
4	831829710	20087	22	COMM	Instructor Incentives	66	0		Average Age		
5	837926805	44325	20	SOCI	Control	47	0		Average Test Score		
6	680763815	45063	18	COMM	Instructor Incentives	96	1		Fraction Passed		
7	252477075	20081	19	SOCI	Control	0	0		Total Observations		
8	323624471	20019	19	ARTH	Instructor Incentives	0	0				
9	205439099	44032	18	COMM	Control	72	1				
10	469048838	20027	20	COMM	Control	48	0				
11	133898507	44019	20	COMM	Control	54	0				
12	408296889	39591	18	COMM	Instructor Incentives	78	1				
13	699523636	27936	18	MATH	Control	43	0				
14	525390693	35167	20	ENGL	Instructor Incentives	84	1				
15	644949942	44213	19	APHY	Control	42	0				

Average Age

- To compute average age in the dataset we just need to use the AVERAGE function and reference column C

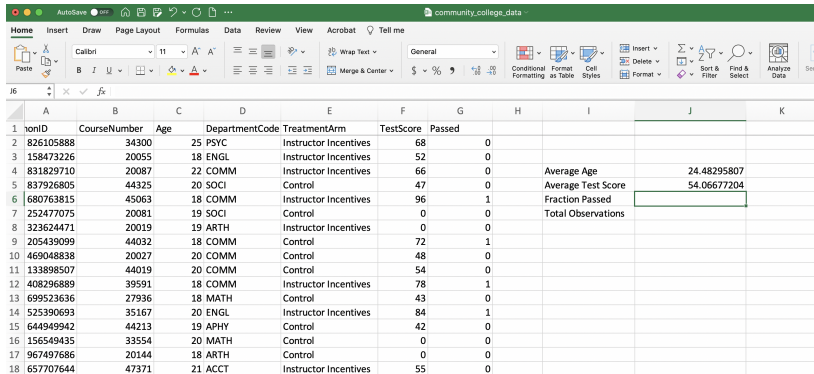


The screenshot shows the Microsoft Excel interface with the 'community_college_data' workbook open. The 'Formulas' tab is active, and the formula bar displays '=AVERAGE(C:C)'. The spreadsheet contains the following data:

	A	B	C	D	E	F	G	H	I	J	K
1	anonID	CourseNumber	Age	DepartmentCode	TreatmentArm	TestScore	Passed				
2	826105888	34300	25	PSYC	Instructor Incentives	68	0				
3	158473226	20055	18	ENGL	Instructor Incentives	52	0				
4	831829710	20087	22	COMM	Instructor Incentives	66	0		Average Age	=AVERAGE(C:C)	
5	837926805	44325	20	SOCI	Control	47	0		Average Test Score		
6	680763815	45063	18	COMM	Instructor Incentives	96	1		Fraction Passed		
7	252477075	20081	19	SOCI	Control	0	0		Total Observations		
8	323624471	20019	19	ARTH	Instructor Incentives	0	0				
9	205439099	44032	18	COMM	Control	72	1				
10	469048838	20027	20	COMM	Control	48	0				
11	133898507	44019	20	COMM	Control	54	0				
12	408296889	39591	18	COMM	Instructor Incentives	78	1				
13	699523636	27936	18	MATH	Control	43	0				
14	525390693	35167	20	ENGL	Instructor Incentives	84	1				
15	644949942	44213	19	APHY	Control	42	0				
16	156549435	33554	20	MATH	Control	0	0				
17	967497686	20144	18	ARTH	Control	0	0				
18	657707644	47371	21	ACCT	Instructor Incentives	55	0				

Average Test Score

- Average test score is completely analogous to age, but we now reference column F



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K
1	nonID	CourseNumber	Age	DepartmentCode	TreatmentArm	TestScore	Passed				
2	826105888	34300	25	PSYC	Instructor Incentives	68	0				
3	158473226	20055	18	ENGL	Instructor Incentives	52	0				
4	831829710	20087	22	COMM	Instructor Incentives	66	0		Average Age	24.48295807	
5	837926805	44325	20	SOCI	Control	47	0		Average Test Score	54.06677204	
6	680763815	45063	18	COMM	Instructor Incentives	96	1		Fraction Passed		
7	252477075	20081	19	SOCI	Control	0	0		Total Observations		
8	323624471	20019	19	ARTH	Instructor Incentives	0	0				
9	205439099	44032	18	COMM	Control	72	1				
10	469048838	20027	20	COMM	Control	48	0				
11	133898507	44019	20	COMM	Control	54	0				
12	408296889	39591	18	COMM	Instructor Incentives	78	1				
13	699523636	27936	18	MATH	Control	43	0				
14	525390693	35167	20	ENGL	Instructor Incentives	84	1				
15	644949942	44213	19	APHY	Control	42	0				
16	156549435	33554	20	MATH	Control	0	0				
17	967497686	20144	18	ARTH	Control	0	0				
18	657707644	47371	21	ACCT	Instructor Incentives	55	0				

Fraction Passed

- ▶ The average of a variable X is

$$\bar{X} = \frac{X_1 + \dots + X_N}{N}$$

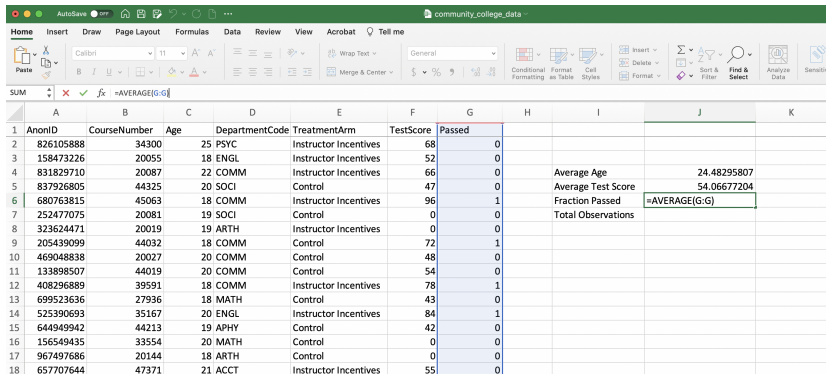
- ▶ where N is the number of observations, X_1 is the first observation and X_N is the N^{th} observation
- ▶ If X is a binary variable that is equal to 1 or zero, then the average will be the fraction of individual's with a 1

$$\bar{X} = \frac{X_1 + \dots + X_N}{N} = \frac{\text{Number of observations} = 1}{\text{Total Observations}}$$

- ▶ Since our variable “Passed” is equal to 1 if passed and zero otherwise, we can calculate the fraction who passed as the average of our “Passed” variable

Fraction Passed

- Fraction passed can be computed as average of column G



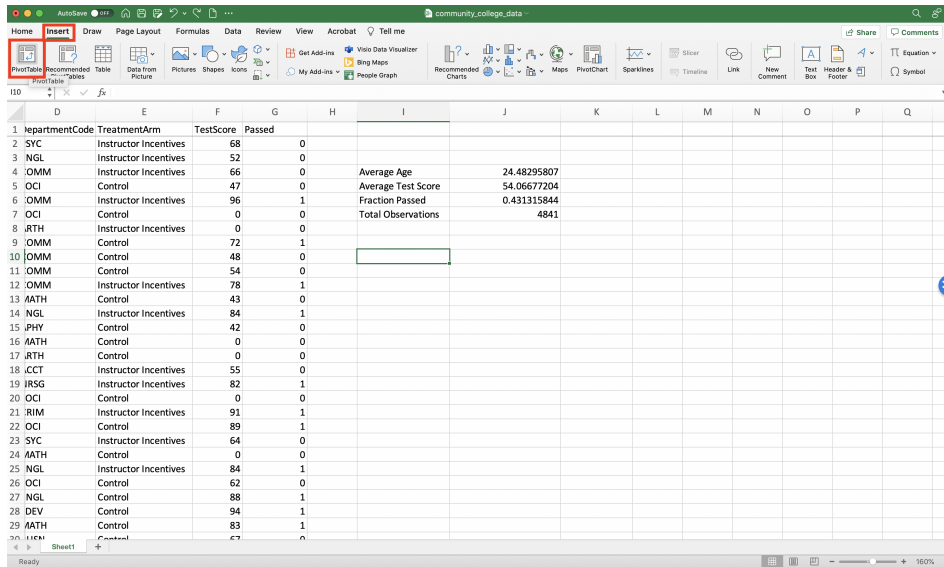
The screenshot shows an Excel spreadsheet with a dataset in columns A through F and summary statistics in columns I and J. The formula bar shows the formula `=AVERAGE(G:G)` entered in cell J3.

	A	B	C	D	E	F	G	H	I	J	K
1	AnonID	CourseNumber	Age	DepartmentCode	TreatmentArm	TestScore	Passed				
2	826105888	34300	25	PSYC	Instructor Incentives	68	0				
3	158473226	20055	18	ENGL	Instructor Incentives	52	0				
4	831829710	20087	22	COMM	Instructor Incentives	66	0		Average Age	24.48295807	
5	837926805	44325	20	SOCI	Control	47	0		Average Test Score	54.06677204	
6	680763815	45063	18	COMM	Instructor Incentives	96	1		Fraction Passed	<code>=AVERAGE(G:G)</code>	
7	252477075	20081	19	SOCI	Control	0	0		Total Observations		
8	323624471	20019	19	ARTH	Instructor Incentives	0	0				
9	205439099	44032	18	COMM	Control	72	1				
10	469048838	20027	20	COMM	Control	48	0				
11	133898507	44019	20	COMM	Control	54	0				
12	408296889	39591	18	COMM	Instructor Incentives	78	1				
13	699523636	27936	18	MATH	Control	43	0				
14	525390693	35167	20	ENGL	Instructor Incentives	84	1				
15	644949942	44213	19	APHY	Control	42	0				
16	156549435	33554	20	MATH	Control	0	0				
17	967497686	20144	18	ARTH	Control	0	0				
18	657707644	47371	21	ACCT	Instructor Incentives	55	0				

Pivot Tables

- ▶ Let's now look at our main focus of this section: does providing instructor's incentives increase test scores?
 - ▶ We need to retrieve average test scores by treatment arm
 - ▶ This is possible using functions, but much more convenient using pivot tables

- ▶ To insert a pivot table, go to the Insert tab and click “Insert Pivot Table”



The screenshot shows the Microsoft Excel interface with the 'Insert' tab selected. The 'PivotTable' icon in the 'Tables' group is highlighted with a red box. The spreadsheet contains data for a study with columns for DepartmentCode, TreatmentArm, TestScore, and Passed. Summary statistics are calculated in columns I and J.

	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	DepartmentCode	TreatmentArm	TestScore	Passed										
2	SYC	Instructor Incentives	68	0										
3	NGL	Instructor Incentives	52	0										
4	OMM	Instructor Incentives	66	0		Average Age	24.48295807							
5	OCI	Control	47	0		Average Test Score	54.06677204							
6	OMM	Instructor Incentives	96	1		Fraction Passed	0.431315844							
7	OCI	Control	0	0		Total Observations	4841							
8	IRTH	Instructor Incentives	0	0										
9	OMM	Control	72	1										
10	OMM	Control	48	0										
11	OMM	Control	54	0										
12	OMM	Instructor Incentives	78	1										
13	AATH	Control	43	0										
14	NGL	Instructor Incentives	84	1										
15	PHY	Control	42	0										
16	AATH	Control	0	0										
17	IRTH	Control	0	0										
18	CCT	Instructor Incentives	55	0										
19	IRSG	Instructor Incentives	82	1										
20	OCI	Control	0	0										
21	RIM	Instructor Incentives	91	1										
22	OCI	Control	89	1										
23	SYC	Instructor Incentives	64	0										
24	AATH	Control	0	0										
25	NGL	Instructor Incentives	84	1										
26	OCI	Control	62	0										
27	NGL	Control	88	1										
28	DEV	Control	94	1										
29	AATH	Control	83	1										
30	UCH	Control	67	0										

- To select a table or range, click on a column (for example D), hold down, and then dragging to another column (for example G). Whatever variable you want to include in the table needs to be in this range

The screenshot shows a Microsoft Excel spreadsheet with a data table. The table has columns labeled C through G, with headers: DepartmentCode, TreatmentArm, TestScore, and Passed. The data rows are numbered 1 through 29. A green selection box highlights the range from column D to column G, rows 1 to 29. A 'Create PivotTable' dialog box is open, showing the 'Table/Range' field with the selected range. The 'Use an external data source' option is selected. The 'Choose where to place the PivotTable' section shows 'Existing worksheet' selected, with the 'Table/Range' field set to 'Sheet1!\$I\$10'.

	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		DepartmentCode	TreatmentArm	TestScore	Passed										
2		25	PSYC	Instructor Incentives	68	0									
3		18	ENGL	Instructor Incentives	52	0									
4		22	COMM	Instructor Incentives	66	0									
5		20	SOCI	Control	47	0									
6		18	COMM	Instructor Incentives	96	1									
7		19	SOCI	Control	0	0									
8		19	ARTH	Instructor Incentives	0	0									
9		18	COMM	Control	72	1									
10		20	COMM	Control	48	0									
11		20	COMM	Control	54	0									
12		18	COMM	Instructor Incentives	78	1									
13		18	MATH	Control	43	0									
14		20	ENGL	Instructor Incentives	84	1									
15		19	APHY	Control	42	0									
16		20	MATH	Control	0	0									
17		18	ARTH	Control	0	0									
18		21	ACCT	Instructor Incentives	55	0									
19		46	NRSF	Instructor Incentives	82	1									
20		21	SOCI	Control	0	0									
21		22	CRIM	Instructor Incentives	91	1									
22		26	SOCI	Control	89	1									
23		19	PSYC	Instructor Incentives	64	0									
24		18	MATH	Control	0	0									
25		20	ENGL	Instructor Incentives	84	1									
26		20	SOCI	Control	62	0									
27		19	ENGL	Control	88	1									
28		20	SDEV	Control	94	1									
29		20	MATH	Control	83	1									

Summary statistics (rows 30-32):

	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
30															
31															
32															

- Once you select a range, Excel will automatically fill in the Table/Range

The screenshot shows the Microsoft Excel interface with a data table and a 'Create PivotTable' dialog box. The data table has columns C through G, with headers: DepartmentCode, TreatmentArm, TestScore, and Passed. The data rows are numbered 1 through 29. The 'Create PivotTable' dialog box is open, showing the 'Table/Range' as 'Sheet1!\$D:\$G' and the 'Table/Range' as 'Sheet1!\$I:\$I0'. The dialog box also has options for 'Choose the data that you want to analyze' and 'Choose where to place the PivotTable'.

	C	D	E	F	G	H	I
1		DepartmentCode	TreatmentArm	TestScore	Passed		
2	25	PSYC	Instructor Incentives	68	0		
3	18	ENGL	Instructor Incentives	52	0		
4	22	COMM	Instructor Incentives	66	0		
5	20	SOCI	Control	47	0		
6	18	COMM	Instructor Incentives	96	1		
7	19	SOCI	Control	0	0		
8	19	ARTH	Instructor Incentives	0	0		
9	18	COMM	Control	72	1		
10	20	COMM	Control	48	0		
11	20	COMM	Control	54	0		
12	18	COMM	Instructor Incentives	78	1		
13	18	MATH	Control	43	0		
14	20	ENGL	Instructor Incentives	84	1		
15	19	APHY	Control	42	0		
16	20	MATH	Control	0	0		
17	18	ARTH	Control	0	0		
18	21	ACCT	Instructor Incentives	55	0		
19	46	NRSRG	Instructor Incentives	82	1		
20	21	SOCI	Control	0	0		
21	22	CRIM	Instructor Incentives	91	1		
22	26	SOCI	Control	89	1		
23	19	PSYC	Instructor Incentives	64	0		
24	18	MATH	Control	0	0		
25	20	ENGL	Instructor Incentives	84	1		
26	20	SOCI	Control	62	0		
27	19	ENGL	Control	88	1		
28	20	SDEV	Control	94	1		
29	20	MATH	Control	83	1		

- By default the table will be placed in whichever cell you clicked last (in my case this is cell I10)

- ▶ Excel has drawn a blank Pivot table. On the right we can see the new panel “PivotTable Fields” which we will use to build our Pivot table

The screenshot shows the Microsoft Excel interface with a workbook named "community_college_data". The data is organized in a table with columns D through N. The first four columns (D, E, F, G) contain the source data: DepartmentCode, TreatmentArm, TestScore, and Passed. A PivotTable has been created in the range I10:M15, with the PivotTable Fields task pane open on the right. The task pane shows the available fields: DepartmentCode, TreatmentArm, TestScore, and Passed. The PivotTable itself is currently blank, displaying the message "To build a report, choose fields from the PivotTable Field List".

	D	E	F	G	H	I	J	K	L	M	N
1	DepartmentCode	TreatmentArm	TestScore	Passed							
2	PSYC	Instructor Incentives	68	0							
3	ENGL	Instructor Incentives	52	0							
4	COMM	Instructor Incentives	66	0		Average Age	24.48295807				
5	SOCI	Control	47	0		Average Test Score	54.06677204				
6	COMM	Instructor Incentives	96	1		Fraction Passed	0.431315844				
7	SOCI	Control	0	0		Total Observations	4841				
8	ARTH	Instructor Incentives	0	0							
9	COMM	Control	72	1							
10	COMM	Control	48	0							
11	COMM	Control	54	0							
12	COMM	Instructor Incentives	78	1							
13	MATH	Control	43	0							
14	ENGL	Instructor Incentives	84	1							
15	APHY	Control	42	0							
16	MATH	Control	0	0							
17	ARTH	Control	0	0							
18	ACCT	Instructor Incentives	55	0							
19	NRSG	Instructor Incentives	82	1							
20	SOCI	Control	0	0							
21	CRIM	Instructor Incentives	91	1							
22	SOCI	Control	89	1							
23	PSYC	Instructor Incentives	64	0							
24	MATH	Control	0	0							
25	ENGL	Instructor Incentives	84	1							
26	SOCI	Control	62	0							
27	ENGL	Control	88	1							
28	SDEV	Control	94	1							
29	MATH	Control	83	1							
30	PSYC	Control	67	0							

- ▶ We want each **row** in our pivot table to be a treatment arm. So click the box next to TreatmentArm to put it into the pivot table

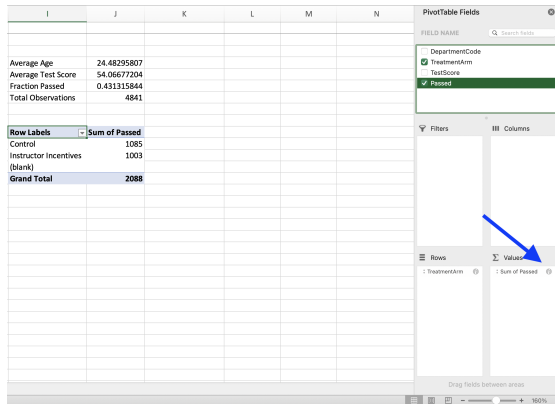
The screenshot shows an Excel worksheet with a PivotTable and the PivotTable Fields task pane. The PivotTable has 'Row Labels' and 'Grand Total' in the first column. The PivotTable Fields task pane shows the following fields:

Field Name
DepartmentCode
☒ TreatmentArm
☐ TestScore
☐ Passed

The task pane also shows the 'Rows' section with 'TreatmentArm' listed. The 'Columns' section is empty. The 'Values' section is empty. The 'Filters' section is empty. The task pane has a search bar and a 'Drag fields between areas' button at the bottom.

- ▶ If excel does not automatically place TreatmentArm in the row panel, you can simply click and drag it to the appropriate panel

- In the Values panel, we want pass rates. We will compute these pass rates from the variable “Passed”



The screenshot shows an Excel PivotTable and the PivotTable Fields task pane. The PivotTable is structured as follows:

Row Labels	Sum of Passed
Control	1085
Instructor Incentives (blank)	1003
Grand Total	2088

The PivotTable Fields task pane on the right shows the following configuration:

- Field Name:** Search fields
- Fields:** DepartmentCode, TreatmentArm, TestScore, Passed (selected)
- Filters:**
- Columns:**
- Rows:** TreatmentArm
- Values:** Sum of Passed (indicated by a blue arrow)

Drag fields between areas

- By default Excel is displaying the “SUM” by TreatmentArm (i.e. number of individuals that passed, not the fraction that passed)

- To change the displayed statistic click the “i” next to the “Values” label and then select “Average”

The screenshot shows an Excel spreadsheet with a PivotTable. The PivotTable has 'Row Labels' and 'Sum of Passed' as the only field. The data is as follows:

Row Labels	Sum of Passed
Control	1085
Instructor Incentives (blank)	1003
Grand Total	2088

A 'PivotTable Field' dialog box is open, showing the 'Source field: Passed' and 'Field name: Average of Passed'. The 'Summarize by' tab is selected, and the 'Average' option is highlighted in the list. The 'OK' button is green.

The 'PivotTable Fields' task pane on the right shows the following configuration:

- FIELD NAME:** Search fields
- DepartmentCode** (unchecked)
- TreatmentArm** (checked)
- TestScore** (unchecked)
- Passed** (checked)
- Filters:** (empty)
- Columns:** (empty)
- Rows:** TreatmentArm
- Values:** Sum of Passed

At the bottom of the task pane, it says 'Drag fields between areas'.

- ▶ Now we've reached our goal. A table that shows average pass rates by Treatment Arm

Row Labels	Average of Passed
Control	0.400369004
Instructor Incentives	0.470671046
(blank)	
Grand Total	0.431315844

- ▶ Giving instructors' incentives increases pass rates by 7 percentage points! That is a big impact!

Small Aside – Percent Impact vs. Percentage Point

- ▶ The treatment increased pass rates by 7 **percentage points** (from a 40 percent pass rate to a 47 percent rate)
- ▶ This is **not** the same as saying the treatment increased pass rates by 7 percent
- ▶ The **percent increase** in pass rates for the treatment relative to control is given by

$$\frac{\text{Pass Rate Treatment} - \text{Pass Rate Control}}{\text{Pass Rate Control}} = \frac{0.471 - 0.400}{0.400} = 0.1775 \quad (1)$$

- ▶ So the treatment pass rate is 17.75 percent higher relative to the control
 - ▶ This mistake can be misleading! Increasing something from 1 percent to 2 percent is a 1 percentage point increase but a 100 percent increase

Interpreting Our Effects

- ▶ The relatively large increase in pass rates due to financial incentives is an important finding!
- ▶ Prior work on “pay-for-performance” in the United States has shown limited effectiveness in K-12 schools (See Neal (2011))
- ▶ But this is the first study to show financial incentives can be an effective tool in the Community College setting
 - ▶ Lesson: the context of the treatment matters! Just because a treatment is effective in one setting doesn't mean it will be effective in another (or vice-versa)